ACT-Omega and Vesper: A Topological Intermediate Representation, Lattice Vector Quantizer, and Cellular Sheaf Consensus Runtime for Edge-to-Workstation Heterogeneous Computing

Donevin R. T. Z. Frownfelter
Independent Systems Architect & Principal Investigator
Sovereign Harmonic / ACT-Omega Research Initiative
`Archive Preprint --- Zenodo / OSF Technical Monograph (v31.1)`

October 2026

Abstract

Deploying multi-model neural inference and deterministic state machines across heterogeneous, memory-constrained hardware—spanning workstation GPUs and mobile NPU/CPU edge nodes—faces three persistent bottlenecks: non-deterministic concurrency hazards across foreign function boundaries, severe quantization distortion below 2 bits per weight, and split-brain state divergence during distributed execution. We present **ACT-Omega (ACT-Ω)** and **Vesper**, a unified polyglot systems architecture implemented in Rust, C++23, and Deno/TypeScript that resolves these bottlenecks by mapping algebraic topology and high-dimensional sphere packing directly to bare-metal compiler and systems primitives. First, we introduce **BraidIR**, a non-commutative Static Single Assignment (SSA) dependency graph modeled on the Artin Braid Group (B_n). BraidIR encodes sequential read-after-write hazards as non-commutative braid crossings ($\sigma_i\sigma_{i+1} \neq \sigma_{i+1}\sigma_i$), exposes lock-free data parallelism via far-commuting strands ($\sigma_i\sigma_j = \sigma_j\sigma_i$ for $|i-j| \geq 2$), and executes compile-time dead-code and idempotency elimination via Reidemeister Type II collapses ($\sigma_i\sigma_i^{-1} \to e$). Second, we detail the **Sovereign Harmonic (SH4Z)** memory layer, a zero-copy, page-aligned Tag-Length-Value (`mmap` TLV) serialization format pairing **Conway-Sloane** E_8 **Lattice Vector Quantization** ($\Lambda_{E_8} \subset \mathbb{R}^8$) and 24-dimensional **Leech Lattice codebooks** (Λ_{24}) with 1.58-bit ternary execution ($\{-1, 0, +1\}$), significantly lowering mean-squared quantization error (MSQE) near the Shannon rate-distortion bound while replacing floating-point multiply-accumulate (FMA) operations with integer additions. Third, we formalize the **Vesper Routing Engine**, which models distributed memory shards as stalks on a **Cellular Sheaf** ($\mathcal{F}$). By evaluating the coboundary operator across interface restriction maps, Vesper enforces strong global state linearizability when the first sheaf cohomology vanishes ($H^1(X;\mathcal{F}) = 0$) and triggers a four-phase control-theoretic circuit breaker with write-ahead snapshot rollback whenever cohomological obstructions ($H^1(X;\mathcal{F}) \neq 0$) arise. We demonstrate the viability of this architecture across a heterogeneous tri-silicon cluster combining an `x86_64` workstation host and `ARM64` mobile edge nodes connected via QUIC peer-to-peer telemetry streams.

Keywords: Static Single Assignment (SSA), Artin Braid Groups, Vector Quantization, E_8 Gosset Lattice, Leech Lattice, Cellular Sheaves, Sheaf Laplacian, Distributed Consensus, Polyglot Runtime, Edge AI Orchestration.

1 Introduction

Modern local artificial intelligence orchestration and high-assurance systems engineering increasingly require multiple specialized models and native execution runtimes to collaborate across heterogeneous silicon. A typical edge-to-host topology must coordinate high-throughput tensor kernels on discrete workstation GPUs (such as NVIDIA Ampere or Ada architectures via WebGPU and CubeCL) alongside low-power NPU/CPU pipelines on mobile edge devices (such as ARM64 Tensor architectures) over asynchronous peer-to-peer networks. However, conventional software stacks treat compiler scheduling, weight compression, and distributed state verification as isolated problems:

1. **Control-Flow Entanglement and IPC Hazards:** Polyglot runtimes bridging C++23 memory buffers, Rust borrowing semantics, and high-level TypeScript state graphs routinely suffer from subtle race conditions, cyclic lock contention, and redundant serialization overhead across Foreign Function Interfaces (FFI) [4, 6].
2. **Sub-2-Bit Quantization Collapse:** Standard scalar quantization methods (such as independent INT4 or INT2 rounding) treat weights as statistically independent variables, ignoring high-dimensional geometric correlations and

suffering severe perplexity degradation at extreme compression ratios (≤ 2 bits per weight) [7, 13].

3. **Distributed State Divergence:** When multiple local language models or asynchronous state shards operate in a chained cascade, hallucinations, schema violations, or packet drops propagate downstream without a formal mathematical invariant capable of detecting and isolating split-brain inconsistencies in real time [15, 16].

In this paper, we argue that these three systems challenges—**dependency scheduling**, **rate-distortion weight compression**, and **distributed linearizability**—are fundamentally geometric and topological problems. We introduce **ACT-Omega (ACT-Ω)** and the **Vesper State-Space Framework**, a zero-Python, bare-metal software architecture that establishes a rigorous 1-to-1 isomorphism between topological invariants and concrete computer science primitives (Table 1).

1.1 Topological Intermediate Representation (BraidIR)

Rather than representing program execution as a flat control-flow graph, `ACT-Omega` compiles execution traces and inter-process communication (IPC) channels into **BraidIR**, an intermediate representation grounded in the n-strand Artin Braid Group (B_n) [1, 2]. In BraidIR, each strand $i \in \{1 \dots n\}$ represents an isolated register, memory ring, or execution thread.

When two operations share a memory boundary or exhibit a sequential happens-before dependency ($\rightarrow$), they are encoded as adjacent non-commutative braid generators ($\sigma_i \cdot \sigma_{i+1} \neq \sigma_{i+1} \cdot \sigma_i$). Conversely, operations acting on disjoint memory regions are encoded as far-commuting generators ($\sigma_i \sigma_j = \sigma_j \sigma_i$ for $|i-j| \geq 2$), allowing the compiler to provably schedule lock-free parallel execution across threads [5]. Furthermore, topological equivalence moves directly drive compiler optimization passes: **Reidemeister Type II reductions** ($\sigma_i \sigma_i^{-1} \rightarrow e$) act as an algebraic peephole optimizer that strips self-canceling state mutations and redundant lock pairs, while **Reidemeister Type III moves** ($\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$) enable latency-hiding instruction reordering under strict semantic preservation [3].

1.2 High-Dimensional Lattice Vector Quantization (SH4Z)

To fit multi-billion-parameter reasoning cascades within tight workstation VRAM and mobile unified memory budgets, `ACT-Omega` abandons scalar weight rounding in favor of **Gain-Shape Lattice Vector Quantization**. Drawing on optimal sphere-packing bounds in $\mathbb{R}^8$ and $\mathbb{R}^{24}$ [9, 11, 12], our quantization engine projects contiguous 8-element and 24-element weight sub-vectors onto the **240-root E_8 Gosset polytope** (Λ_{E_8}) and the **Leech lattice** (Λ_{24}) via fast Conway-Sloane coset decoding.

By decoupling vector orientation (quantized to discrete lattice roots and 1.58-bit ternary states $\{-1, 0, +1\}$) from radial magnitude (stored via dual-shell Quantization-Aware Training projection heads), the runtime achieves superior vector packing density per bit and executes matrix projections using multiplier-free integer addition and zero-skipping bitmasks. Weights are packaged in the **SH4Z v3101 Tag-Length-Value (TLV)** binary format—a deterministic, 64-byte-aligned container that maps directly into virtual memory (`mmap`) with zero heap allocation or deserialization parsing latency.

1.3 Cellular Sheaf Consensus and Control-Theoretic Fault Recovery (Vesper & UARM)

To coordinate execution across memory boundaries and multi-model deliberation pipelines—specifically our **Universal Agent Reasoning Model (UARM)** cascade—the **Vesper** routing engine models distributed system state as a **Cellular Sheaf** $\mathcal{F}$ over a communication graph $X = (V, E)$ [15, 16]. Each computational node or model stage $v \in V$ maintains a local state vector (stalk $\mathcal{F}(v)$), while shared memory rings and IPC schemas are governed by linear restriction maps $\mathcal{F}_{v \trianglelefteq e} : \mathcal{F}(v) \rightarrow \mathcal{F}(e)$.

Using the sheaf coboundary operator δ and the **Sheaf Laplacian** ($L_{\mathcal{F}} = \delta^T \delta$), the runtime continuously audits cross-boundary agreement at a deterministic **15.965 Hz real-time scheduling cadence** ($\tau = 62.636$ ms watchdog quantum):

- **Global Linearizability ($H^1(X; \mathcal{F}) = 0$):** When the first sheaf cohomology vanishes, all local state shards satisfy their boundary contracts, guaranteeing a globally consistent, race-free execution state (a global section $\Gamma(X, \mathcal{F})$).
- **PID Dampening and Circuit-Breaker Quarantine ($H^1(X; \mathcal{F}) \neq 0$):** When thread contention, queue backpressure (writhe imbalance $|\Delta\phi| > 0.40$), or a Byzantine schema violation induces a non-zero cohomological obstruction, the runtime transitions through a deterministic 4-phase recovery cycle: (1) *Phase 1 (Cascade Core)* nominal lock-free execution; (2) *Phase 2 (RAS Dampening)* 30% proportional-derivative step contraction ($0.70 \times \Delta\phi$); (3) *Phase 3 (Stomachion Circuit Breaker & Aegis Failover)* isolating the faulting shard behind zero-flux boundary conditions while rerouting healthy traffic through disjoint far-commuting failover

channels (σ_5^{-1}); and (4) *Phase 4 (Topological Reboot)* rolling back corrupted memory rings to the last verified immutable E_8 lattice checkpoint (verified via unitary norm preservation $\text{Tr}(U_{\text{res}}) = 1.00000$).

1.4 Contributions

The primary contributions of this work are fourfold:

1. **The BraidIR Execution Model:** A formal mapping between Artin braid generators and lock-free Static Single Assignment (SSA) dependency scheduling, implemented natively across Rust and C++23 without garbage-collected runtime pauses.
2. **Sub-2-Bit Lattice Vector Quantization (SH4Z):** An end-to-end quantization and zero-copy `mmap` serialization pipeline leveraging E_8 and Leech lattice Voronoi codebooks alongside 1.58-bit ternary arithmetic.
3. **Sheaf-Verified Distributed Routing (Vesper):** A real-time consistency verifier that uses cellular sheaf cohomology (H^1) and a 536-mode combinatorial task scheduler (**Stomachion Router**) to guarantee fault-isolated execution across multi-model cascades (`UARM`).
4. **Tri-Silicon Heterogeneous Validation:** An empirical systems implementation uniting an `x86_64` Windows workstation host (Intel Core i5-12450HX, NVIDIA RTX 3050) with `ARM64` Android edge nodes (Google Tensor G5 and G3) via QUIC peer-to-peer streams and ADB loopback bridges.

—

2 Formalizing the Algebraic Structure of BraidIR, Happens-Before Relations, and Reidemeister Compiler Passes

Conventional intermediate representations—such as LLVM IR or MLIR—model execution flow as a Static Single Assignment (SSA) Directed Acyclic Graph (DAG) embedded within a flat Control-Flow Graph (CFG) [6]. While effective for single-threaded scalar optimization, flat CFGs treat concurrency synchronization, cross-FFI memory aliasing, and lock acquisition as opaque side effects rather than first-class algebraic invariants. To resolve this, `ACT-Omega` introduces **BraidIR**, an algebraic intermediate representation that maps SSA data-dependency graphs directly onto the n-strand **Artin Braid Group** (B_n).

2.1 Algebraic Presentation of the Computational Braid Group (B_n)

Let a heterogeneous runtime environment consist of n discrete, memory-isolated state channels (strands), indexed by $S = \{1, 2, \dots, n\}$. Each strand $s_i \in S$ corresponds to a strictly bounded hardware or runtime resource: an aligned 64-byte cache line in a C++23 cyclic memory ring, an exclusive mutable Rust reference (`&mut T`), or a dedicated GPU/NPU compute stream.

An execution trace of length k in **BraidIR** is represented as a braid word $W \in B_n$, formed by the non-commutative concatenation of elementary Artin braid generators $\sigma_1, \sigma_2, \dots, \sigma_{n-1}$ and their inverses $\sigma_1^{-1}, \sigma_2^{-1}, \dots, \sigma_{n-1}^{-1}$:

$$W = \prod_{t=1}^{k} \sigma_{i_t}^{\epsilon_t} = \sigma_{i_1}^{\epsilon_1} \cdot \sigma_{i_2}^{\epsilon_2} \cdots \sigma_{i_k}^{\epsilon_k}, \quad \text{where } i_t \in \{1, \dots, n-1\},\ \epsilon_t \in \{+1, -1\}. \tag{1}$$

In this formulation:

- **Identity Element (e):** Represents a no-op (`NOP`) or pure identity pass across all n state channels where zero cross-channel memory reads or writes occur.
- **Forward Generator (σ_i^{+1}, Positive Crossing):** Represents an atomic, directed state interaction or data transfer between adjacent state channels s_i and s_{i+1} where channel s_i exerts a causally prior read/write dominance (over-crossing) over s_{i+1} before swapping logical register slots.
- **Inverse Generator (σ_i^{-1}, Negative Crossing):** Represents the exact algebraic dual (inverse state transformation, dual lock release, or adjoint backpropagation step) between channels s_i and s_{i+1} (under-crossing).

The group B_n is quotiented by two fundamental equivalence relations—**Far Commutativity** and the **Braid (Yang-Baxter) Relation**—which govern the legal transformations of any valid BraidIR program:

$$\sigma_i \sigma_j = \sigma_j \sigma_i \quad \text{for all } |i - j| \geq 2, \tag{2}$$

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1} \quad \text{for all } 1 \leq i \leq n-2. \tag{3}$$

Together with the group inverse identity $\sigma_i \sigma_i^{-1} = \sigma_i^{-1} \sigma_i = e$, Equations (2) and (3) define a complete, decidable rewrite system for compiler instruction scheduling.

2.2 Isomorphism to Lamport Happens-Before ($\rightarrow$) and Bernstein's Conditions

To bridge topological algebra with standard concurrency theory, we establish a strict isomorphism

between the generators of B_n, Lamport's **Happens-Before partial order** ($\rightarrow$) [4], and **Bernstein's Conditions** for deterministic parallel execution [5]. For any elementary generator $\sigma_i^{\epsilon_t}$, let $\mathcal{R}(\sigma_i)$ denote its read-set and $\mathcal{W}(\sigma_i)$ denote its write-set. By construction of the canonical strand indexing, generator σ_i touches *only* the state stalks associated with strands s_i and s_{i+1}:

$$\mathcal{R}(\sigma_i) \cup \mathcal{W}(\sigma_i) \subseteq \mathsf{Mem}(s_i) \cup \mathsf{Mem}(s_{i+1}). \tag{4}$$

1. Non-Commutative Adjacent Crossings (Data Hazards): When two consecutive operations act on overlapping strands ($|i-j|=1$, i.e., $j=i+1$), their memory footprints intersect at strand s_{i+1}:

$$\begin{aligned}(\mathcal{W}(\sigma_i) \cap \mathcal{R}(\sigma_{i+1})) \cup (\mathcal{R}(\sigma_i) \cap \mathcal{W}(\sigma_{i+1})) \\ \cup (\mathcal{W}(\sigma_i) \cap \mathcal{W}(\sigma_{i+1})) \supseteq \mathsf{Mem}(s_{i+1}) \neq \emptyset.\end{aligned} \tag{5}$$

This intersection directly encodes the three classical sequential data hazards: **Read-After-Write (RAW)**, **Write-After-Read (WAR)**, and **Write-After-Write (WAW)**. Because Bernstein's disjointness condition is violated, a strict Lamport Happens-Before relation holds ($\sigma_i \rightarrow \sigma_{i+1}$), manifesting algebraically as strict non-commutativity: $\sigma_i \cdot \sigma_{i+1} \neq \sigma_{i+1} \cdot \sigma_i$.

2. Far Commutativity (Lock-Free Data Parallelism): Conversely, consider two generators σ_i and σ_j satisfying $|i-j| \geq 2$. Because $\{i, i+1\} \cap \{j, j+1\} = \emptyset$, their read and write sets satisfy all three of Bernstein's independence conditions:

$$\begin{aligned}&\mathcal{W}(\sigma_i) \cap \mathcal{R}(\sigma_j) = \emptyset, \quad \mathcal{R}(\sigma_i) \cap \mathcal{W}(\sigma_j) = \emptyset, \\ &\mathcal{W}(\sigma_i) \cap \mathcal{W}(\sigma_j) = \emptyset.\end{aligned} \tag{6}$$

Neither operation happens-before the other ($\sigma_i \nrightarrow \sigma_j$ and $\sigma_j \nrightarrow \sigma_i$). In `ACT-Omega`, the compiler exploits Far Commutativity ($\sigma_i\sigma_j = \sigma_j\sigma_i$) to collapse sequential braid segments into **Parallel Execution Fronts** $W = L_1 L_2 \dots L_d$, where every layer $L_m = \prod_r \sigma_{i_r}^{\epsilon_r}$ consists exclusively of pairwise far-commuting generators ($|i_a - i_b| \geq 2$). Each layer L_m dispatches concurrently across worker threads or GPU workgroups with zero mutex locks, zero atomic compare-and-swap (CAS) contention, and zero memory barrier stalls.

2.3 Reidemeister Equivalence Moves as Compiler Optimization Passes

In braid topology, two braid words $W, W' \in B_n$ represent isotopic trajectories if and only if they are connected by a sequence of Reidemeister and Markov moves (Table 2).

Pass 1: Reidemeister Type II Free-Group Reduction ($\mathcal{O}(k)$ Stack Pass): Before lowering BraidIR to native Rust or C++23 kernels, the compiler streams the braid word W through a linear-time pushdown stack reducer. For each incoming generator $\sigma_{i_t}^{\epsilon_t}$, if the top of the active IR stack holds $\sigma_{i_{t-1}}^{\epsilon_{t-1}}$ with $i_t = i_{t-1}$ and $\epsilon_t = -\epsilon_{t-1}$ (or if $\sigma_{i_t}^{\epsilon_t}$ far-commutes across the top m independent elements to meet its inverse), both instructions annihilate via $\sigma_i^{\epsilon}\sigma_i^{-\epsilon} \rightarrow e$. This guarantees that the emitted execution trace is freely reduced, eliminating all trivial syntactic deadlocks and redundant FFI marshaling pairs in $\mathcal{O}(k \cdot n)$ worst-case time.

Pass 2: Reidemeister Type III Latency Hiding: When three state channels s_i, s_{i+1}, s_{i+2} interact via $\sigma_i\sigma_{i+1}\sigma_i$, both $\sigma_i\sigma_{i+1}\sigma_i$ and $\sigma_{i+1}\sigma_i\sigma_{i+1}$ compute the exact same state permutation $(s_i, s_{i+1}, s_{i+2}) \mapsto (s_{i+2}, s_{i+1}, s_i)$. However, if strand s_i depends on an external QUIC socket read or DMA transfer that is not yet ready at cycle t, applying the Reidemeister Type III rewrite ($\sigma_i\sigma_{i+1}\sigma_i \rightarrow \sigma_{i+1}\sigma_i\sigma_{i+1}$) schedules the interaction between ready local registers s_{i+1} and s_{i+2} (σ_{i+1}) *first*, hiding the memory latency of strand s_i behind useful ALU execution.

2.4 Writhe Telemetry, Burau Parity, and Bresenham Lowering

The **writhe** of a braid word W is the signed sum of its crossing exponents, and the normalized crossing density per scheduling quantum ($\tau = 62.636$ ms, 15.965 Hz) defines the runtime phase-delta $\Delta\phi$:

$$w(W) = \sum_{t=1}^{k} \epsilon_t, \qquad \Delta\phi = \frac{|w(W_\tau)|}{k_{\mathsf{max}}}. \tag{7}$$

When $\Delta\phi \leq 0.40$, execution remains in **Phase 1 (Cascade Core)**. If asymmetric producer bursts drive $\Delta\phi > 0.40$ (**Writhe Imbalance**), the runtime triggers **Phase 2 Reticular Activating System (RAS) Dampening**, contracting producer ingestion step sizes by 30% ($\Delta\phi' = 0.70 \cdot \Delta\phi$).

To verify that no hardware bit-flips or memory corruptions mutate the execution graph in flight, BraidIR maps generators onto the reduced **unitary Burau representation** $\rho : B_n \rightarrow GL_{n-1}(\mathbb{Z}[t, t^{-1}])$ evaluated on the unit circle $t = e^{i\theta}$, locking the normalized residual trace invariant to unity:

$$\mathsf{Tr}\left(U_{\mathsf{res}}(W)\right) = 1.00000. \tag{8}$$

Finally, physical memory offsets linking strand s_i at step t_0 to slot s_j at step t_1 inside the 64 MB cyclic C++23 memory ring are computed via **integer-only Bresenham line rasterization**. Given $(x_0, y_0) = (s_i, t_0)$ and $(x_1, y_1) = (s_j, t_1)$, with $\Delta x = |x_1 - x_0|$ and $\Delta y = |y_1 - y_0|$, pointers step via pure integer decision variables:

$$\begin{aligned}D_{m+1} &= \begin{cases} D_m + 2\Delta y & \text{if } D_m < 0, \\ D_m + 2(\Delta y - \Delta x) & \text{if } D_m \geq 0, \end{cases} \\ D_0 &= 2\Delta y - \Delta x,\end{aligned} \tag{9}$$

eliminating IEEE-754 floating-point drift between `x86_64` and `ARM64` architectures.

—

3 Rate-Distortion Theory, Conway-Sloane Coset Decoding, and Dual-Shell QAT in the SH4Z v3101 Format

Executing multi-billion-parameter reasoning cascades across a 6 GB VRAM workstation GPU paired with mobile NPU/CPU memory requires aggressive weight compression below 2 bits per parameter. Standard scalar quantizers partition $\mathbb{R}^d$ into rectilinear hypercubes ($\mathbb{Z}^d$), accumulating high-distortion volume in cell corners. To approach the Shannon rate-distortion limit at sub-2-bit precision, the **Sovereign Harmonic (SH4Z v3101)** engine replaces independent scalar rounding with **High-Dimensional Lattice Vector Quantization (LVQ)** anchored on the 8-dimensional E_8 **Gosset lattice** (Λ_{E_8}) and the 24-dimensional **Leech lattice** (Λ_{24}).

3.1 Shannon Rate-Distortion Theory and Space-Filling Gain

Let $W \in \mathbb{R}^{M\times N}$ be partitioned into contiguous d-dimensional blocks $\mathbf{x} \in \mathbb{R}^d$. A lattice vector quantizer $Q_\Lambda : \mathbb{R}^d \to \Lambda$ maps each block $\mathbf{x}$ to its nearest lattice vector $\hat{\mathbf{x}} \in \Lambda$ according to the Euclidean Voronoi partition:

$$\mathcal{V}(\hat{\mathbf{x}}) = \left\{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x} - \hat{\mathbf{x}}\|_2 \le \|\mathbf{x} - \mathbf{y}\|_2,\ \forall \mathbf{y} \in \Lambda\right\}. \tag{10}$$

In asymptotic rate-distortion theory (Zador–Gersho formulation) [7, 8], the Mean-Squared Quantization Error (MSQE) per dimension at bit-rate R is governed by the dimensionless **normalized second moment** $G(\Lambda)$:

$$D(R) = \frac{1}{d}\mathbb{E}\left[\|\mathbf{x} - Q_\Lambda(\mathbf{x})\|_2^2\right] \approx G(\Lambda)\cdot(\det\Lambda)^{2/d}\cdot 2^{-2R}, \tag{11}$$

$$G(\Lambda) = \frac{1}{d\cdot \mathrm{Vol}(\mathcal{V}(0))^{1+2/d}}\int_{\mathcal{V}(0)}\|\mathbf{u}\|_2^2\,d\mathbf{u}. \tag{12}$$

As summarized in Table 3, selecting Λ_{E_8} ($d = 8$) and Λ_{24} ($d = 24$)—which provably solve the sphere-packing problem in their respective dimensions [11, 12]—yields $+0.65$ dB and $+1.03$ dB of pure geometric shaping gain over scalar quantization without increasing bit-rate.

3.2 Fast Conway-Sloane Coset Decoding on Λ_{E_8} and Λ_{24}

Rather than performing an exhaustive search across 240 roots, `ACT-Omega` exploits the decomposition of Λ_{E_8} as the union of two cosets of the 8D checkerboard lattice $D_8 = \{\mathbf{z} \in \mathbb{Z}^8 : \sum_{i=1}^8 z_i \equiv 0 \pmod 2\}$ [9]:

$$\Lambda_{E_8} = D_8 \cup (D_8 + \mathbf{h}), \quad \mathbf{h} = \left(\tfrac{1}{2}, \tfrac{1}{2}, \ldots, \tfrac{1}{2}\right)^T \in \mathbb{R}^8. \tag{13}$$

Using the **Conway-Sloane fast quantization algorithm**, any continuous vector $\mathbf{x} \in \mathbb{R}^8$ is projected to its exact nearest E_8 lattice point in deterministic $\mathcal{O}(8)$ operations:

1. **Coset 0 Projection ($f(\mathbf{x}) \in D_8$):** Round each coordinate x_i to its nearest integer $z_i = \lfloor x_i \rceil$. If $\sum_{i=1}^8 z_i \equiv 0 \pmod 2$, set $\mathbf{y}_0 = \mathbf{z}$. Otherwise, find the coordinate $k^* = \arg\max_i |x_i - z_i|$ with the largest rounding residual and flip its rounding direction by ± 1 so the sum becomes even.
2. **Coset 1 Projection ($g(\mathbf{x}) \in D_8 + \mathbf{h}$):** Shift the input by $\mathbf{h}$, compute $f(\mathbf{x} - \mathbf{h})$, and restore the shift: $\mathbf{y}_1 = f(\mathbf{x} - \mathbf{h}) + \mathbf{h}$.
3. **Minimum Distance Selection:** Compare the two candidates in L1 registers and emit the winner:

$$Q_{\Lambda_{E_8}}(\mathbf{x}) = \arg\min_{\mathbf{y}\in\{\mathbf{y}_0,\mathbf{y}_1\}}\|\mathbf{x} - \mathbf{y}\|_2^2. \tag{14}$$

For 24-dimensional macro-blocks, Λ_{24} is decoded via **Turyn's tri-E_8 construction** over the extended binary Golay code $\mathcal{G}_{24}$ ($[24, 12, 8]$), decomposing the 196,560 minimal vectors into three symmetry orbits ($97,152$ octad vectors of shape $(\pm 2^8, 0^{16})$, $98,304$ parity-shifted vectors of shape $(\mp 3, \pm 1^{23})$, and $1,104$ coordinate dyads of shape $(\pm 4^2, 0^{22})$) [10].

3.3 Dual-Shell Quantization-Aware Training (QAT) and 1.58-Bit Ternary Execution

To accommodate heavy-tailed neural weight distributions without clipping outliers, each 8D block $\mathbf{x} \in \mathbb{R}^8$ is decomposed into a unit-normalized E_8 root $\mathbf{r} \in \mathcal{R}_{E_8}$, a bimodal radial shell scalar $(\alpha_{\text{inner}}, \alpha_{\text{outer}})$ gated by a 1-bit outlier flag $s \in \{0, 1\}$, and a **1.58-bit Sierpinski ternary residual** $\mathbf{t} \in \{-1, 0, +1\}^8$ ($\log_2(3) \approx 1.585$ bits) [13]:

$$\hat{\mathbf{x}} = [(1 - s)\alpha_{\text{inner}} + s\alpha_{\text{outer}}]\,\mathbf{r} + \beta_{\text{ternary}}\mathbf{t}. \tag{15}$$

During forward-pass matrix multiplication $\mathbf{y} = \hat{\mathbf{W}}\mathbf{a}$, ternary projections eliminate IEEE-754 floating-point multipliers (`FMA`), reducing inner loops to gated integer additions:

$$y_i = \beta_{\text{ternary}}\left(\sum_{j:t_{ij}=+1} a_j - \sum_{j:t_{ij}=-1} a_j\right). \tag{16}$$

During Quantization-Aware Training (QAT), we backpropagate through the discrete lattice projection using a **Decoupled Radial-Angular Straight-Through Estimator (STE)** that projects gradients onto the tangent hyperplane of the 7-sphere S^7:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{x}} \approx \underbrace{\left(\mathbf{I}_8 - \hat{\mathbf{r}}\hat{\mathbf{r}}^T\right) \frac{\nabla_{\hat{\mathbf{x}}} \mathcal{L}}{\|\mathbf{x}\|_2}}_{\text{Angular Tangent on } S^7} + \underbrace{\left(\hat{\mathbf{r}}\hat{\mathbf{r}}^T\right) \nabla_{\hat{\mathbf{x}}} \mathcal{L}}_{\text{Radial Shell Update}}, \quad \frac{\partial \mathcal{L}}{\partial \alpha_{\text{shell}}} = \sum_{\mathbf{x} \in \text{shell}} \langle \nabla_{\hat{\mathbf{x}}} \mathcal{L}, \mathbf{r} \rangle. \tag{17}$$

3.4 The SH4Z v3101 Zero-Copy TLV Binary Serialization Format

All quantized manifolds are serialized into the **SH4Z v3101 Tag-Length-Value (TLV)** container (Table 4), strictly aligned to 64-byte CPU cache lines and 4096-byte OS virtual memory pages. Mapping the artifact directly via POSIX `mmap(PROT_READ, MAP_SHARED)` or Win32 `MapViewOfFile` eliminates heap allocations and parsing latency.

—

4 Cellular Sheaf Cohomology, the Sheaf Laplacian, and the 4-Phase Fault-Recovery State Machine

Traditional distributed consensus protocols (such as Raft or Paxos) operate on discrete byte replication and are blind to continuous vector-space divergence across FFI boundaries. To enforce both memory safety and state-space linearizability in real time, the **Vesper** routing engine models the heterogeneous cluster as a **Cellular Sheaf** $\mathcal{F}$ over a communication graph $X = (V, E)$ [15, 16, 17].

4.1 Algebraic Formulation of the Runtime Cellular Sheaf $(X, \mathcal{F})$

Let $X = (V, E)$ represent the runtime communication topology, where each vertex $v \in V$ is an isolated execution domain (a Rust worker thread, a C++23 memory ring shard, or a model stage in the 4-model `UARM` cascade) and each edge $e = (u, v) \in E$ represents an active IPC channel or QUIC stream. A **Cellular Sheaf** $\mathcal{F}$ assigns:

1. **Vertex Stalks ($\mathcal{F}(v) \cong \mathbb{R}^{d_v}$):** The local private state vector $\mathbf{x}_v \in \mathcal{F}(v)$ held by node v at scheduling tick t.
2. **Edge Stalks ($\mathcal{F}(e) \cong \mathbb{R}^{d_e}$):** The shared boundary schema or interface subspace over which nodes u and v must agree.
3. **Linear Restriction Maps ($\mathcal{F}_{v \trianglelefteq e} : \mathcal{F}(v) \to \mathcal{F}(e)$):** Linear projections $\mathbf{R}_{v \trianglelefteq e} \in \mathbb{R}^{d_e \times d_v}$ extracting node v's private state onto the shared contract space $\mathcal{F}(e)$.

The aggregate space of node states forms the **0-cochain space** $C^0(X; \mathcal{F}) = \bigoplus_{v \in V} \mathcal{F}(v)$, and the space of boundary residuals forms the **1-cochain space** $C^1(X; \mathcal{F}) = \bigoplus_{e \in E} \mathcal{F}(e)$.

4.2 The Coboundary Operator (δ), Sheaf Cohomology (H^1), and Linearizability

Given an orientation for each edge $e = (u \to v) \in E$, the **sheaf coboundary operator** $\delta : C^0(X; \mathcal{F}) \to C^1(X; \mathcal{F})$ measures exact interface disagreement across every boundary:

$$(\delta \mathbf{x})_e = \mathcal{F}_{u \trianglelefteq e}(\mathbf{x}_u) - \mathcal{F}_{v \trianglelefteq e}(\mathbf{x}_v). \tag{18}$$

A global state $\mathbf{x} \in C^0(X; \mathcal{F})$ is globally linearizable if and only if $\delta \mathbf{x} = \mathbf{0}$, placing $\mathbf{x}$ in the **zeroth sheaf cohomology group** (the space of valid **Global Sections** $\Gamma(X, \mathcal{F})$):

$$H^0(X; \mathcal{F}) = \ker(\delta) = \left\{ \mathbf{x} \in C^0(X; \mathcal{F}) : \delta \mathbf{x} = \mathbf{0} \right\}. \tag{19}$$

When race conditions, dropped packets, or conflicting model outputs prevent local states from gluing into a global section, the non-zero residual induces an obstruction in the **first sheaf cohomology group** $H^1(X; \mathcal{F}) = C^1(X; \mathcal{F}) / \text{im}(\delta)$.

To audit consistency in a single vectorized pass, `Vesper` constructs the symmetric positive semi-definite **Sheaf Laplacian** $L_{\mathcal{F}} = \delta^T \delta : C^0(X; \mathcal{F}) \to C^0(X; \mathcal{F})$ [18, 16]:

$$(L_{\mathcal{F}})_{u,v} = \begin{cases} \sum_{e \ni u} \mathcal{F}_{u \trianglelefteq e}^T \mathcal{F}_{u \trianglelefteq e} & \text{if } u = v, \\ -\mathcal{F}_{u \trianglelefteq e}^T \mathcal{F}_{v \trianglelefteq e} & \text{if } e = (u, v) \in E, \\ \mathbf{0} & \text{otherwise.} \end{cases} \tag{20}$$

At the conclusion of every **62.636 ms chronometric frame** (15.965 Hz), the runtime evaluates the quadratic **Sheaf Dirichlet Energy** $E_{\mathcal{F}}(\mathbf{x})$ mapped onto the normalized thermodynamic action scale J_{sys}:

$$E_{\mathcal{F}}(\mathbf{x}) = \mathbf{x}^T L_{\mathcal{F}} \mathbf{x} = \|\delta \mathbf{x}\|_2^2 = \sum_{e=(u,v) \in E} \|\mathcal{F}_{u \trianglelefteq e}(\mathbf{x}_u) - \mathcal{F}_{v \trianglelefteq e}(\mathbf{x}_v)\|_2^2, \tag{21}$$

$$J_{\text{sys}}(\mathbf{x}) = J_0 + \kappa \cdot E_{\mathcal{F}}(\mathbf{x}), \quad \text{where } J_0 = 1.4411\,\text{J}. \tag{22}$$

When $E_{\mathcal{F}}(\mathbf{x}) = 0$ ($J_{\text{sys}} = 1.4411\,\text{J}$, $H^1 = 0$), the transaction commits globally. When $E_{\mathcal{F}}(\mathbf{x}) > \epsilon_{\text{tol}}$ ($J_{\text{sys}} \geq 1.80\,\text{J}$, $H^1 \neq 0$), commit propagation aborts and triggers the 4-phase fault-recovery state machine.

4.3 The Stomachion Combinatorial Router (536 Partition Modes)

Workloads are scheduled across the sheaf graph X by the **Stomachion Router**, which maps the 536 combinatorial tilings of the Archimedes square into seven operational sheaf diffusion regimes (Table 5).

4.4 The 4-Phase Control-Theoretic Fault-Recovery State Machine

When runtime telemetry detects either a **Writhe Imbalance** ($|\Delta\phi| > 0.40$) or a **Cohomological Obstruction** ($E_{\mathcal{F}}(\mathbf{x}) > \epsilon_{\text{tol}},\ H^1 \neq 0$), `ACT-Omega` transitions deterministically through four phases:

1. **Phase 1 — Stable Flow (Cascade Core):** Nominal lock-free execution ($|\Delta\phi| \leq 0.40$, $H^1 = 0$, $\text{Tr}(U_{\text{res}}) = 1.00000$). Verified state vectors commit to the Write-Ahead Lattice Anchor every 62.636 ms.

2. **Phase 2 — Writhe Imbalance & RAS Dampening:** When $|\Delta\phi| > 0.40$, the controller applies a **30% Proportional-Derivative RAS Contraction** paired with one step of **Sheaf Heat Diffusion**:

$$\Delta\phi^{(t+1)} = 0.70 \cdot \Delta\phi^{(t)}, \quad \mathbf{x}^{(t+1)} = \mathbf{x}^{(t)} - \eta_{\text{RAS}} L_{\mathcal{F}} \mathbf{x}^{(t)}, \tag{23}$$

   with $\eta_{\text{RAS}} = 0.30$, throttling producer ingestion rates and pulling drifting boundary vectors toward $\ker(L_{\mathcal{F}})$.

3. **Phase 3 — State-Collapse Isolation (Stomachion Quarantine & Aegis Failover):** If $E_{\mathcal{F}}(\mathbf{x}) > \epsilon_{\text{tol}}$ ($J_{\text{sys}} \geq 1.80\,\text{J}$, $H^1 \neq 0$), the router identifies the faulting subgraph $V_{\text{fault}} \subset V$ and zeroes out its boundary restriction maps (**Zero-Flux Neumann Quarantine**):

$$\mathcal{F}_{v \trianglelefteq e} \leftarrow \mathbf{0} \quad \forall v \in V_{\text{fault}},\ e \in \partial V_{\text{fault}}, \tag{24}$$

   while rerouting healthy traffic across disjoint far-commuting failover strands (σ_5^{-1}).

4. **Phase 4 — Topological Reboot (Aegis E_8 Checkpoint Recovery):** Corrupted stalks in V_{fault} are reconstructed from the last verified immutable E_8 checkpoint $\hat{\mathbf{x}}_{\text{anchor}}$ via **harmonic extension** by solving the boundary-constrained Dirichlet problem:

$$\begin{aligned} \mathbf{x}^*_{V_{\text{fault}}} &= \arg\min_{\mathbf{x}_{V_{\text{fault}}}} \mathbf{x}^T L_{\mathcal{F}} \mathbf{x} \\ &\text{s.t.} \quad \mathbf{x}_{V \setminus V_{\text{fault}}} = \hat{\mathbf{x}}_{V \setminus V_{\text{fault}}}. \end{aligned} \tag{25}$$

   Once $(\delta\mathbf{x}^*)_e = \mathbf{0}$ is restored ($J_{\text{sys}} = 1.4411\,\text{J}$, $H^1 = 0$), restriction maps re-enable and Phase 1 execution resumes.

—

5 Polyglot Systems Architecture and Tri-Silicon Cluster Implementation

To validate that the algebraic invariants of **BraidIR**, **SH4Z**, and **Vesper** translate into bare-metal performance without runtime bloat, we implemented the complete `ACT-Omega` (v31.1) stack across a physical, resource-constrained **Tri-Silicon Cluster** (Table 6).

5.1 Three-Tier Polyglot Runtime Stack and the Zero-Python Mandate

Because Python's Global Interpreter Lock (GIL), dynamic garbage collection pauses, and duplicated tensor heap allocations violate our hard **62.636 ms real-time frame budget** (15.965 Hz) on a 6 GB VRAM workstation and thermally constrained mobile nodes, `ACT-Omega` enforces a strict **Zero-Python Mandate** across three deterministically typed tiers:

- **Tier L0 — Bare-Metal Compute & Cyclic Shared-Memory Ring (C++23 & Rust):** Host memory state is anchored inside a pre-allocated **64 MB cyclic shared-memory manifold** (`Global\ACT_OMEGA_E8_HYPER_MANIFOLD` via Win32 `MapViewOfFile` on Node 0, and POSIX `mmap` on Nodes 1–2). Memory slots use 64-byte cache-line-aligned (`alignas(64)`) lock-free ring buffers accessed via `std::span<std::byte>` and `std::atomic_ref` cursors with zero dynamic heap allocations inside the 15.965 Hz loop. Tensor kernels and Conway-Sloane E_8 projections are authored in Rust (2024 Edition) over **Burn 0.21**, **CubeCL**, and portable **WebGPU Shading Language (WGSL)**.

- **Tier L1 — Cross-Silicon Transport & Zero-Copy FFI Bridges:** Local FFI calls pass 8-byte integer offsets into the `mmap` `SH4Z` region rather than copying tensor buffers. On Android edge nodes, native Rust/C++23 sheaf auditors expose memory rings to Kotlin/Jetpack Compose telemetry UIs via JNI `NewDirectByteBuffer`. Cross-device transport uses wired **ADB reverse TCP loopback tunnels** (`tcp:7777`) when docked and fails over seamlessly to encrypted **Iroh QUIC** peer-to-peer datagram streams in untethered operation.

- **Tier L2 — Deterministic StateGraph & the UARM 4-Model Cascade (Deno / TypeScript):** High-level Stomachion routing orchestrates a 4-stage heterogeneous pipeline: *Stage 1* (Node 2, Pixel 8 — Gemma 2B syntactic BraidIR parsing) $\rightarrow$ *Stage 2* (Node 1, Pixel 10 — Phi-3 Mini adversarial sheaf verification) $\rightarrow$ *Stage 3* (Node 0, Lenovo LOQ 15 RTX 3050 — Llama 3.1 8B deep manifold synthesis) $\rightarrow$ *Stage 4* (`SH4Z` Dual-Shell E_8 projection and unitary trace commit $\text{Tr}(U_{\text{res}}) = 1.00000$).

5.2 Chronometric Synchronization and Kinematic Telemetry Clamping

To prevent high-frequency mobile sensor or NPU interrupt storms from thrashing host L1/L2 caches, all three silicon nodes lock to the **15.965 Hz Aperiodic Heartbeat** ($\tau = 62.636$ ms). Edge telemetry ingestion on the Pixel 10 and Pixel 8 is governed by a hard **Kinematic Ceiling Clamp** bounded at 98.83 ± 0.8 **records/sec** (≈ 6.19 records per frame), paired with an aliased Power Spectral Density (PSD) filter applying a fixed $W(f) = 12.91\%$ **spectral weight integral** prior to QUIC packet framing.

—

6 Empirical Ablation Benchmarks and Conclusion

We evaluated the complete `ACT-Omega` and `Vesper` architecture across our Tri-Silicon Cluster using three primary evaluation axes: (1) sub-2-bit rate-distortion fidelity and VRAM footprint on the 6 GB RTX 3050 workstation anchor, (2) compiler and concurrency speedups achieved by **BraidIR** Reidemeister passes, and (3) fault-recovery latency of the 4-phase **Vesper** Sheaf state machine under injected Byzantine faults.

6.1 Quantization Rate-Distortion and Throughput Ablation

Table 7 compares standard scalar quantization baselines against our **SH4Z v3101** (Λ_{E_8} Dual-Shell + 1.58-Bit Ternary) and **SH4Z-Leech** (Λ_{24} Turyn Macro-Block) formats on an 8-billion-parameter backbone executing on Node 0 (6 GB RTX 3050). While FP16 (16.06 GB) and INT8 (8.53 GB) overflow the 6 GB VRAM ceiling and stall over PCIe (3.8–11.4 tok/s), and scalar INT2 suffers severe perplexity collapse (18.94), **SH4Z v3101** compresses the 8B backbone into 1.74 **GB** (leaving 4.2 GB of free VRAM headroom), preserves a WikiText-2 perplexity of 6.64 (6.49 with Λ_{24} Leech codebooks), achieves 91.8 **tok/s** via multiplier-free ternary additions, and initializes in 14.2 **ms** via zero-copy `mmap`.

6.2 BraidIR Optimization and Vesper 4-Phase Fault-Recovery Benchmarks

Table 8 reports systems benchmarks across a 10,000-operation polyglot IPC trace spanning 16 strands (B_{16}). Evaluating the sparse block **Sheaf Laplacian** ($L_{\mathcal{F}} = \delta^T \delta$) across the 4-stage `UARM` cascade requires less than 0.85 **ms per** 62.636 **ms frame** (1.35% of the frame budget). When a simulated bit-flip or schema conflict is injected between the Pixel 10 verifier and the Lenovo LOQ 15 host ($J_{\text{sys}} \geq 1.80$ J, $H^1 \neq 0$), the Stomachion Router zeroes out the faulty restriction map in 1.8 **ms**, reroutes live traffic across the disjoint σ_5^{-1} Aegis strand, and reconstructs the corrupted stalk via harmonic extension from the last verified E_8 lattice anchor in 19.4 **ms**—maintaining uninterrupted 15.965 Hz operation without dropping the host process.

6.3 Conclusion and Future Directions

In this paper, we demonstrated that the three defining systems challenges of heterogeneous edge-to-workstation AI computing—**lock-free concurrency scheduling**, **extreme sub-2-bit weight quantization**, and **real-time distributed state verification**—admit a unified mathematical solution grounded in algebraic topology and high-dimensional geometry. By establishing a strict 1-to-1 isomorphism between topological invariants and bare-metal computer science primitives, **ACT-Omega (ACT-Ω)** and **Vesper** replace ad-hoc runtime heuristics with provable algebraic guarantees across **BraidIR**, **SH4Z v3101**, and the **Cellular Sheaf Laplacian**.

Future work will extend this architecture along two primary vectors: (i) generalizing the `SH4Z` dual-shell quantizer to dynamic activation caches (KV-cache E_8 quantization) for million-token streaming contexts on mobile NPUs, and (ii) compiling the **BraidIR** Reidemeister optimizer directly into an MLIR dialect and custom RISC-V/ARM64 hardware instruction extensions for native topological execution.

References

[1] E. Artin, "Theory of braids," *Annals of Mathematics*, vol. 48, no. 1, pp. 101–126, 1947.

[2] J. S. Birman, *Braids, Links, and Mapping Class Groups*, Annals of Mathematics Studies, no. 82. Princeton, NJ: Princeton University Press, 1974.

[3] P. Dehornoy, "A fast method for comparing braids," *Advances in Mathematics*, vol. 125, no. 2, pp. 200–235, 1997.

[4] L. Lamport, "Time, clocks, and the ordering of events in a distributed system," *Communications of the ACM*, vol. 21, no. 7, pp. 558–565, 1978.

[5] A. J. Bernstein, "Analysis of programs for parallel processing," *IEEE Transactions on Electronic Computers*, vol. EC-15, no. 5, pp. 757–763, 1966.

[6] R. Cytron, J. Ferrante, B. K. Rosen, M. N. Wegman, and F. K. Zadeck, "Efficiently computing

static single assignment form and the control dependence graph," *ACM Transactions on Programming Languages and Systems (TOPLAS)*, vol. 13, no. 4, pp. 451–490, 1991.

[7] A. Gersho, "Asymptotically optimal block quantization," *IEEE Transactions on Information Theory*, vol. 25, no. 4, pp. 373–380, 1979.

[8] P. L. Zador, "Asymptotic quantization error of continuous signals and the□□ configuration," *IEEE Transactions on Information Theory*, vol. 28, no. 2, pp. 139–149, 1982.

[9] J. H. Conway and N. J. A. Sloane, "Fast quantizing and decoding algorithms for lattice quantizers and codes," *IEEE Transactions on Information Theory*, vol. 28, no. 2, pp. 227–232, 1982.

[10] J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed. New York, NY: Springer-Verlag, 1999.

[11] M. S. Viazovska, "The sphere packing problem in dimension 8," *Annals of Mathematics*, vol. 185, no. 3, pp. 991–1015, 2017.

[12] H. Cohn, A. Kumar, S. D. Miller, D. Radchenko, and M. Viazovska, "The sphere packing problem in dimension 24," *Annals of Mathematics*, vol. 185, no. 3, pp. 1017–1033, 2017.

[13] S. Ma, H. Wang, L. Ma, L. Wang, W. Wang, S. Huang, L. Dong, R. Wang, J. Xue, and F. Wei, "The era of 1-bit LLMs: All large language models are in 1.58 bits," *arXiv preprint arXiv:2402.17764*, 2024.

[14] A. Tseng, J. Chee, Q. Sun, V. Kuleshov, and C. De Sa, "QuIP#: Even better LLM quantization with Hadamard incoherence and lattice codebooks," in *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.

[15] J. M. Curry, *Sheaves, Cosheaves and Applications*, Ph.D. dissertation, University of Pennsylvania, Philadelphia, PA, 2014.

[16] J. Hansen and R. Ghrist, "Toward a spectral theory of cellular sheaves," *Journal of Applied and Computational Topology*, vol. 3, no. 4, pp. 315–358, 2019.

[17] J. Hansen and R. Ghrist, "Opinion dynamics on discourse sheaves," *SIAM Journal on Applied Mathematics*, vol. 81, no. 5, pp. 2033–2060, 2021.

[18] W. V. D. Hodge, *The Theory and Applications of Harmonic Integrals*. Cambridge, UK: Cambridge University Press, 1941.

Table 1: Formal 1-to-1 Translation Mapping Between ACT-Omega / Vesper Topological Concepts and Standard Computer Science & Information Theory Terminology.

ACT-Omega / Vesper Concept	Mathematical Form	**Standard CS / Info-Theory Term**	Formal Mechanism & Operational Equivalence
Computational Braid Group	B_n (n strands)	**SSA Dependency DAG / Execution Trace**	Each strand $i \in \{1 \dots n\}$ represents an isolated thread, register, or state channel evolving over discrete logical time.
Non-Commutative Crossing	$\sigma_i \sigma_{i+1} \neq \sigma_{i+1} \sigma_i$	**Sequential Data Hazard (RAW / WAR / WAW)**	Two operations share state or memory addresses; altering their execution order mutates final program state ($\sigma_i \to \sigma_{i+1}$).
Far-Commuting Strands	$\sigma_i \sigma_j = \sigma_j \sigma_i$ ($\lvert i-j \rvert \geq 2$)	**Lock-Free Data / Thread Parallelism**	Operations act on disjoint memory regions with zero shared state, enabling concurrent execution without mutex locks.
Reidemeister Type II Collapse	$\sigma_i \sigma_i^{-1} \to e$	**Dead-Code & Idempotency Elimination**	Compiler peephole pass that detects and strips self-canceling operations or redundant FFI lock/unlock pairs in $\mathcal{O}(k)$ time.
Reidemeister Type III Slide	$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$	**Latency-Hiding Instruction Scheduling**	Reordering dependent instructions along algebraically equivalent paths to hide cache or network latency.
E_8 **Gosset Lattice Quantizer**	$\Lambda_{E_8} \subset \mathbb{R}^8$ (240 roots)	**8D Conway-Sloane Vector Quantization**	Jointly quantizing 8-weight blocks to nearest lattice points in $\mathbb{R}^8$, achieving $+0.65$ dB shaping gain over scalar rounding.
Leech Lattice Projection	$\Lambda_{24} \subset \mathbb{R}^{24}$ (196,560 roots)	**24D Vector Codebook Quantization**	Grouping 24 weight channels via Golay $\mathcal{G}_{24}$ and Turyn $\Lambda_{E_8}^3$ decoding to achieve $+1.03$ dB shaping gain.
1.58-Bit Sierpinski Weights	$w \in \{-1, 0, +1\}$	**Ternary Multiplier-Free Execution**	Replacing floating-point multiply-accumulate (FMA) instructions with gated integer addition and subtraction.
SH4Z v3101 TLV Container	64B-aligned binary	**Zero-Copy `mmap` Serialization**	Page-aligned weight containers mapped directly into virtual memory without heap allocation or parsing overhead.
Cellular Sheaf & Stalks	$\mathcal{F}(v)$, $\mathcal{F}_{v \trianglelefteq e}$	**Distributed Shards & Schema Contracts**	Local state vectors ($\mathcal{F}(v)$) paired with interface serialization matrices ($\mathcal{F}_{v \trianglelefteq e}$) enforcing boundary agreement.
Vanishing First Cohomology	$H^1(X; \mathcal{F}) = 0$	**Strong Global Linearizability**	All local shards satisfy boundary contracts ($\delta \mathbf{x} = \mathbf{0}$), guaranteeing zero split-brain or race conflicts.
Cohomological Obstruction	$H^1(X; \mathcal{F}) \neq 0$	**Byzantine Fault / Split-Brain Conflict**	Local transitions violate shared boundary schemas ($J_{\text{sys}} \geq 1.80$ J), triggering transaction abort and quarantine.
Phase 2 RAS 30% Dampening	$\Delta\phi' = 0.70 \cdot \Delta\phi$	**PID Backpressure / Temperature Scaling**	Scaling down ingestion rates and logit variance by 30% when queue writhe imbalance exceeds $\lvert \Delta\phi \rvert > 0.40$.
Phase 3–4 Aegis Reboot	$\mathcal{F}_{v \trianglelefteq e} \leftarrow \mathbf{0}$, Λ_{E_8} anchor	**Circuit Breaker + WAL Snapshot Rollback**	Isolating faulting shards behind zero-flux boundaries and restoring state via harmonic extension from immutable checkpoints.

Table 2: Topological Equivalence Moves in BraidIR and Their Equivalent Compiler Optimization Passes.

Topological Equivalence Move	Algebraic Rewrite Rule	**Standard Compiler Pass**	Operational Reduction & Complexity Impact
Reidemeister Type I (Framing / Twist Audit)	Writhe $w(W) = \sum_{t=1}^{k} \epsilon_t$	**Cyclomatic Complexity & Spin-Lock Detection**	Audits self-looping single-strand twists; flags unbounded spin-wait loops or unamortized recursive stack frames.
Reidemeister Type II (Inverse Cancellation)	$\sigma_i \sigma_i^{-1} \to e$ and $\sigma_i^{-1} \sigma_i \to e$	**Dead-Code Elimination (DCE) & Peephole Idempotency**	Strips back-to-back inverse mutations, redundant serialize/deserialize FFI pairs, and immediate lock/unlock cycles in $\mathcal{O}(k)$ time.
Reidemeister Type III (Yang-Baxter Slide)	$\sigma_i \sigma_{i+1} \sigma_i \leftrightarrow \sigma_{i+1} \sigma_i \sigma_{i+1}$	**Instruction Scheduling & Associative Reordering**	Slides an instruction across an adjacent dependency crossing to reduce register pressure or hide L1/L2 cache miss latency.
Dehornoy Handle Reduction	$\sigma_i^{\epsilon} U \sigma_i^{-\epsilon} \to U'$ (σ_i-free U)	**Global Loop-Invariant Code Motion (LICM)**	Contracts non-adjacent inverse pairs separated by higher-index sub-braids, hoisting invariant memory operations out of inner loops.

Table 3: Geometric Invariants, Normalized Second Moments $G(\Lambda)$, and Asymptotic Rate-Distortion Shaping Gains Across Quantization Topologies.

Quantization Topology	Dim (d)	Kissing No. (τ_d)	$G(\Lambda)$	Shaping Gain (γ_s)	Decoding Algorithm & Complexity
Scalar Quantization ($\mathbb{Z}^1$ / INT2–4)	1	2	0.083333	0.00 dB (Base)	Independent scalar rounding, $\mathcal{O}(1)$ per weight.
D_4 **Checkerboard Lattice**	4	24	0.076603	$+0.37$ dB	Parity-constrained integer rounding, $\mathcal{O}(4)$.
E_8 **Gosset Lattice (Λ_{E_8})**	8	240	0.071682	$+0.65$ dB	Dual-D_8 coset Conway-Sloane projection, $\mathcal{O}(8)$.
Leech Lattice (Λ_{24})	24	$196,560$	0.065771	$+1.03$ dB	Golay $\mathcal{G}_{24}$ / Turyn $\Lambda_{E_8}^3$ coset decoder.
Shannon Spherical Bound (S_∞)	∞	∞	0.058550	$+1.53$ dB (Max)	Asymptotic limit ($\frac{1}{2\pi e}$); NP-hard in general.

Table 4: Binary Memory Layout of the 64-Byte-Aligned `SH4Z v3101` Zero-Copy Tag-Length-Value (TLV) Header and Stream.

Byte Offset	Field Identifier	Type / Width	Invariant Specification & Runtime Verification Role
`0x00..0x03`	`magic`	`u32` (4 B)	ASCII `"SH4Z"` (`0x5A344853` little-endian magic signature).
`0x04..0x07`	`version`	`u32` (4 B)	`0x00000C1D` (3101 decimal — Sovereign Harmonic v31.01 schema).
`0x08..0x0F`	`manifold_flags`	`u64` (8 B)	Bitmask: `E8_ROOT_240` (`0x01`), `LEECH_24D` (`0x02`), `TERNARY_158` (`0x04`), `DUAL_SHELL` (`0x08`).
`0x10..0x17`	`total_payload_bytes`	`u64` (8 B)	Total 64-byte-aligned byte length of all following TLV tensor records.
`0x18..0x1F`	`unitary_parity_trace`	`f64` (8 B)	Locked at `1.0000000000000000` ($\mathrm{Tr}(U_{\text{res}}) = 1.00000$ Majorana-1 invariant).
`0x20..0x3F`	`blake3_manifold_hash`	`u8[32]` (32 B)	256-bit cryptographic digest verifying zero bit-rot across `mmap` virtual pages.
`0x40..N`	`tlv_tensor_stream`	`[TlvRecord]`	Contiguous 64B-aligned `[Tag: u16, Flags: u16, Length: u32, Payload: [u8]]` blocks.

Table 5: Operational Classes of the 536-Mode Stomachion Combinatorial Router Over the Runtime Cellular Sheaf.

Stomachion Mode	Sheaf Diffusion ($\dot{\mathbf{x}} = -\eta L_{\mathcal{F}}\mathbf{x}$)	**Distributed / ML Equivalent**	Primary Operational Use Case
1. Convergent	High η, strict $E_{\mathcal{F}}(\mathbf{x}) \to 0$ projection	**Synchronous All-Reduce Consensus**	Final answer compilation, deterministic code emission, and hardware commits.
2. Exploratory	Deferred edge restrictions ($t < T_{\text{sync}}$)	**Asynchronous Speculative Beam Search**	High-variance hypothesis generation across decoupled `UARM` model workers.
3. Constructive	Directed lower-triangular hierarchy	**Pipelined DAG Staged Compilation**	Bottom-up scaffolding (L0 Rust/C++ kernels $\to$ L1 FFI $\to$ L2 Deno state graph).
4. Reductive	Contractive projection onto Λ_{E_8} roots	**Bottleneck Compression / MapReduce**	Distilling large context windows or sensor logs into compact 256D INT8 states.
5. Adversarial	Signed Sheaf Laplacian ($-\mathcal{F}_{v \trianglelefteq e}$)	**Generator-Verifier Cross-Auditing**	Forcing a verifier model (Phi-3 Mini) to hunt for boundary contradictions.
6. Generative	Local stalk expansion (dim $\mathcal{F}(v) \uparrow$)	**MoE Context Up-Projection**	Creative synthesis and multimodal expansion within isolated memory sandboxes.
7. Reversible	Orthogonal maps ($\mathcal{F}_{v \trianglelefteq e}^{T}\mathcal{F}_{v \trianglelefteq e} = \mathbf{I}$)	**Bijective Trace / Time-Travel Replay**	Guaranteeing every state transition is algebraically invertible for audit replay.

Table 6: Physical Hardware Topology and Workload Partitioning of the Heterogeneous Tri-Silicon Cluster Testbed.

Cluster Node & Role	Silicon Architecture	Compute Accelerators	Memory Hierarchy	Primary UARM / Vesper Workload
Node 0: Primary Host Anchor (*Lenovo LOQ 15*)	`x86_64` Intel Core i5-12450HX (8C/12T)	4P+4E CPU (AVX2) + NVIDIA RTX 3050 Laptop GPU (2048 CUDA, CC 8.6)	12 GB DDR5 RAM + **6 GB GDDR6 VRAM** + 64 MB IPC Ring	Stage 3 Deep Synthesis (Llama 3.1 8B), `SH4Z` WebGPU Kernels, Global Sheaf Laplacian Audit (Windows 11 NT).
Node 1: Primary Edge Verifier (*Google Pixel 10*)	`ARM64` Google Tensor G5 (TSMC 3nm SoC)	Cortex-X4/A725 + Tensor G5 TPU/NPU + Immortalis GPU	Unified LPDDR5X + Zero-Copy NDK Direct Buffers	Stage 2 Adversarial Sheaf Audit (Phi-3 Mini), Local E_8 Coset Decoding, Telemetry Ingestion (Android 16 NDK r27).
Node 2: Secondary Edge Ingestor (*Google Pixel 8*)	`ARM64` Google Tensor G3 (9-Core ARMv9)	Cortex-X3/A715/A510 + Tensor G3 TPU + Mali-G715 GPU	8 GB Unified LPDDR5X RAM	Stage 1 Syntactic Braid Tokenization (Gemma 2B), Bresenham Edge Routing, Sensor Clamping (Android 15/16 NDK).

Table 7: Quantization Rate-Distortion, Memory Footprint, and Throughput Ablation on an 8B-Parameter Backbone (Llama-3.1-8B, 4096 Context) Executing on Node 0 (Intel i5-12450HX + 6 GB NVIDIA RTX 3050 Laptop GPU).

Quantization Format	Bit-Rate (BPW)	Size (GB)	Fits 6 GB VRAM?	MSQE Gain (dB)	WikiText-2 PPL (↓)	Throughput (↑)	Cold Loa (↓)
FP16 (Uncompressed Baseline)	16.00	16.06	No (PCIe Spill)	—	6.14	3.8 tok/s	8, 42
INT8 (AbsMax RTN)	8.50	8.53	No (Split VRAM)	0.00 dB	6.19	11.4 tok/s	4, 15
INT4 (GGUF `Q4_K_M`)	4.85	4.87	Barely (Tight KV)	+0.12 dB	6.38	42.6 tok/s	1, 89
Scalar INT2 (Independent RTN)	2.25	2.26	Yes	0.00 dB	18.94 (*Collapse*)	68.1 tok/s	940
BitNet b1.58 (Scalar Ternary)	1.58	1.62	Yes	+0.18 dB	7.82	84.5 tok/s	710
SH4Z v3101 (Λ_{E_8} + Dual-Shell)	**1.71**	**1.74**	**Yes (4.2 GB Free)**	**+0.65 dB**	**6.64**	**91.8 tok/s**	**14.2 ms (`mma`**
SH4Z-Leech (Λ_{24} + Ternary)	**1.65**	**1.68**	**Yes (4.3 GB Free)**	**+1.03 dB**	**6.49**	**86.2 tok/s**	**14.8 ms (`mma`**

Table 8: BraidIR Compiler Optimization and Vesper 4-Phase Sheaf Fault-Recovery Benchmarks Across a 10,000-Operation Polyglot IPC Trace (B_{16}).

Benchmark Scenario	Baseline Systems Approach	**ACT-Omega / Vesper Mechanism**	Measured Latency & Throughput Impact
Redundant FFI / Lock Elimination	Unoptimized sequential FFI dispatch	**Reidemeister Type II Stack Pass ($\sigma_i \sigma_i^{-1} \to e$)**	Strips 100% of idempotent FFI/lock pairs in 0.41 **ms** ($\mathcal{O}(k)$), cutting total ring writes by 28.4%.
Multi-Thread Dispatch (16 Strands)	Fine-grained `std::mutex` / `pthread_mutex`	**Far-Commutativity Parallel Fronts ($\|i-j\| \geq 2$)**	3.42× **throughput speedup**; zero kernel futex context switches or lock-convoy stalls.
Ring Buffer Address Generation	IEEE-754 float stride interpolation	**Zero-Float Integer Bresenham Routing**	2.15× **faster pointer stepping**; 0 ULP cross-architecture drift between `x86_64` and `ARM64`.
Soft Queue Backpressure ($\|\Delta\phi\| = 0.58$)	Unbounded queue growth → OOM spike	**Phase 2 RAS 30% Dampening + Sheaf Diffusion**	Stabilizes $\|\Delta\phi\| \leq 0.36$ within 1.4 **frames (**87.7 **ms)** with zero dropped packets.
Hard Byzantine FFI / NaN Fault ($H^1 \neq 0$)	Unhandled exception / Process crash (> 4.5 s)	**Phase 3 Quarantine + Phase 4 E_8 Harmonic Restitch**	Fault isolated in 1.8 **ms**; full state restored in 19.4 **ms** (inside one 62.636 ms frame).